2.9 - divergence to $\pm\infty$

end - formal definition of $\lim_{x \to a} f(x) = L$.

Math 421

Monday, November 3

Announcements

- Homework 8 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
  - Email me & fill out the form if you have a conflict and need to take the makeup.
  - Sign up for McBurney exam if applicable.

Set $\delta = \min\{1, \frac{\varepsilon}{k}\}$

Activity – Critique the Proof

Theorem. $\lim_{x \to 1} x^2 + x + 1 = 3$.

δ must be a constant – cannot include x.

False Proof. Fix $\epsilon > 0$. Set $\boxed{\delta = \frac{\epsilon}{|x+2|}}$. Assume $0 < |x-1| < \delta$.

$|x-1| < 1$

Then $|(x^2 + x + 1) - 3| = |x^2 + x - 2| = |(x-1)(x+2)|$

find a bound for $|x+2|$

$$= |x-1||x+2| < \delta \cdot \boxed{|x+2|} = \frac{\epsilon}{|x+2|} \cdot |x+2| = \epsilon.$$

k

$\delta = \frac{\varepsilon}{k}$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to 1} x^2 + x + 1 = 3$.

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

$``= a"$ (under 1) $``= L"$ (above 3)

Discussion.

Algorithm for quadratic polynomials:

1. Factor out $|x - a|$ from $|f(x) - L|$

   $|x-1|$ $\quad |(x^2+x+1)-3|$

   $|x^2 + x - 2| = |(x-1)(x+2)|$

   $|x-1|\boxed{|x+2|}$

2. Deal with the extra factor

   a) Write in terms of $x - a$: $\quad |x+2| = |(x-1)+3|$

   b) Use triangle inequality: $\quad \le |x-1| + 3 \quad < 2+3 = 5$

   c) Assume a bound of 1: $\quad < 1 + 3 = 4$

   $|x-1| < 1$ ← choice. or 2

3. Set $\delta = \min\{1, \epsilon/\text{answer in c)}\}$

   $\delta = \min\{1, \epsilon/4\}$

   $\delta = \min\{2, \epsilon/5\}$

$|x-1| < 1$

$|x-1| < \frac{\varepsilon}{4}$

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

Proof. Fix $\varepsilon > 0$. Set $\delta = \min\{1, \frac{\varepsilon}{4}\}$. Assume $0 < |x-1| < \delta$

$= |(x-1)(x+2)| =$

Then $|(x^2+x+1) - 3| = |x^2+x-2| = |x-1||x+2|$

$= |x-1||(x-1)+3| \leq |x-1|(|x-1|+3)$

$< |x-1|(1+3) < \frac{\varepsilon}{4} \cdot 4 = \varepsilon.$

Since $\varepsilon > 0$ was arbitrary, $\lim_{x \to 1} x^2 + x + 1 = 3$.

Activity – Fill in the Blanks

$|x-3|<1$

$-1 < x-3 < 1$

$2 < x < 4$

$4 < \boxed{x+2 < 6}$

$2 < |x+2| < 6$

Thm. $\lim_{x \to 3} x^2 - x = 6.$

Proof. Fix $\epsilon > 0$. Set $\delta = \min\{1, \underline{\ \varepsilon/6\ }\}$. Assume $0 < |x-3| < \delta$.

Then $|x^2 - x - 6| = \underline{|(x-3)(x+2)| = |x-3||x+2|}$

$\underline{= |x-3||(x-3)+5| \leq |x-3|(1+5)}$

$\underline{< \frac{\varepsilon}{6} \cdot 6}$ $= \cancel{<}\ \epsilon.$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to 3} x^2 - x = 6.$

Example – $\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$

Discussion. We want $\left| x^2 \sin\frac{1}{x} - 0 \right| < \varepsilon$ whenever $|x-0| < \delta$

$\left| x^2 \sin\frac{1}{x} \right| = |x^2| \left| \sin\frac{1}{x} \right| \le |x^2| \cdot 1 < \varepsilon \qquad \delta = \sqrt{\varepsilon}$

Proof.

Functional Limits – via Sequences

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S. Then $\boxed{\lim_{x \to a} f(x) = L}$ if for every sequence (x_n) in S satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, we have $\boxed{\lim_{n \to \infty} f(x_n) = L}$.

Example: Let $f(x) = x^2$, $\lim_{x \to 0} x^2 = 0$.

$(x_n) = \left(\frac{1}{n}\right)$ $\quad \lim \frac{1}{n} = 0 = a$

$\left(f(x_n)\right) = \left(\left(\frac{1}{n}\right)^2\right)$ $\quad \lim \left(\frac{1}{n}\right)^2 = 0 = L$